Info Fusion - Sage Lab Synthesis:
Improving antibiotic resistance prediction and interpretability with mutli-modal input

Jackson Michaels

Project Outline

Background - Bio

Genetic Overview:

- Subject is Mycobacterium tuberculosis, reference sequence H37Rv
  - 4.41 million **base pairs** aka **nucleotide pairs**
- per antibiotic we have annotated **Loci**, regions of interest relating to said medication. Typically are in genic region but can be intergenic
  - roughly 7,500 base pairs long
- sets of three nucleotides **(codons)** are encoded as **amino acids**
- amino acids come together to construct **proteins**
- generally, **one protein per gene** for TB
- Regions can be **genic** (inside a gene) or **intergenic** (outside a gene)
  - the latter are of particular interest because they regulate and promote gene expression

Background - Project

Goal: take existing work from SAGE lab on antibiotic resistance prediction in TB and improve on performance and interpretability by incorporating multimodal insights from Info Fusion lab

Task Details:

- Original input data is presented as set of loci, 5 X ~7,500 vector of base pairs for single drug
  - these loci can be genic regions or intergenic
  - typically one hot encoded *(A, C, T, G, gap)*
  - covers 2-3 genes and intergenic regions
- task is to predict resistance to eleven antibiotic drugs
  - multi-drug input is concatenated single-drug inputs

Modalities - DNA

Top Choices:

- Regulatory regions
- more than 2-3 proteins

DNA-Derived:

- one-hot array / ref-alt encoding
- foundation model embedding
- K-mer frequency (31-mers used in literature)

```
>NC_000962.3 NC_000962.3:314309..314854 (-
strand) class=CDS length=546

GTGCACACCCAGGTACACACGGCCCGCCTGGTCCAC
ACCGCCGATCTTGACAGCGAGACCCGCCAGGACATC
CGTCAGATGGTCACCGGCGCGTTTGCCGGTGACTTCA
CCGAGACCGACTGGGAGCACACGCTGGGTGGGATGC
ACGCCCTGATCTGGCATCACGGGGCGATCATCGCGC
ATGCCGCGGTGATCCAGCGGCGACTGATCTACCGCG
GCAACGCGCTGCGCTGCGGGTACGTCGAAGGCGTTG
CGGTGCGGGCGGACTGGCGGGGCCAACGCCTGGTG
AGCGCGCTGTTGGACGCCGTCGAGCAGGTGATGCGC
GGCGCTTACCAGCTCGGAGCGCTCAGTTCCTCGGCG
CGGGCCCGCAGACTGTACGCCTCACGCGGCTGGCTG
CCCTGGCACGGCCCGACATCGGTACTGGCACCAACC
GGTCCAGTCCGTACACCCGATGACGACGGAACGGTG
TTCGTCCTGCCCATCGACATCAGCCTGGACACCTCGG
CGGAGCTGATGTGCGATTGGCGCGCGGGCGACGTCT
GGTAA
```

Example Loci: Rv0262c

Modalities - Regulatory

Regulatory Regions:

- high-value, mechanistically-motivated modality that current CNN baselines under-use
- handled as separate modality from DNA

Description:

- regulates the downstream gene(s) it precedes
  - e.g. promoters / operators
- regulates expression or lack of expression for previous genes

Motivation:

- Resistance can arise without changing the protein at all, by a promoter mutation that over-expresses the gene

Modalities - Protein

Description:

- Translation from DNA sequence
- effectively 3-mers

Representations:

- one-hot array / ref-alt encoding of amino-acids
- foundation model embedding

Notes:

- another top choice is to simply include more than 2-3 in the inference

```
>Mycobacterium tuberculosis

H37Rv|Rv0262c|aac

VHTQVHTARLVHTADLDSETRQDIRQMVTGAFAGDFT
ETDWEHTLGGMHALIWHHGAIIAHAAVIQRRLIYRGNA
LRCGYVEGVAVRADWRGQRLVSALLDAVEQVMRGAYQ
LGALSSSARARRLYASRGWLPWHGPTSVLAPTGPVRTP
DDDGTVFVLPIDISLDTSAELMCDWRAGDVW
```

Example Loci: Rv0262c, amino-acid sequence

Modalities - 3D Structure

Description:

- spatially describe distance of base-pairs in folded protein

Representations:

- distance matrix (e.g. alpha fold)
- Density vector (mine)
- Shrake-Rupley algorithm

Notes:

- only useful if two proteins interact

Modalities - Molecular Characteristics

Description:

- Chemical properties of molecules in protein separate from function
- Represents how individual molecules behave and interact
- Used in *(Kulkarni et al)*

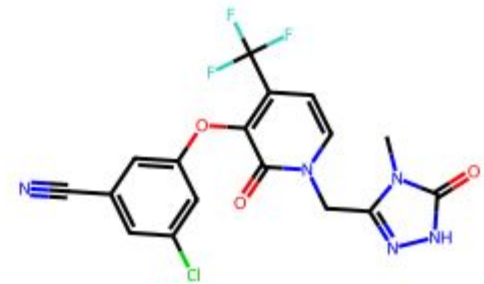
Features:

- Molecular weight
- Isoelectric point
- Hydro / Lipophilicity
- Refractivity

Notes:

- derived from RDKit Library

RDKit
Open-Source Cheminformatics and Machine Learning



```
getMolDescriptors(doravirine)

{'MaxEStateIndex': 13.412553309006833,
 'MinEStateIndex': -4.871620672188628,
 'MaxAbsEStateIndex': 13.412553309006833,
 'MinAbsEStateIndex': 0.045220418860841605,
 'qed': 0.6914051268589834,
 'MolWt': 425.754,
...
'fr_thiophene': 0,
 'fr_unbrch_alkane': 0,
 'fr_urea': 0}
```

Modalities - Lineage

Description:

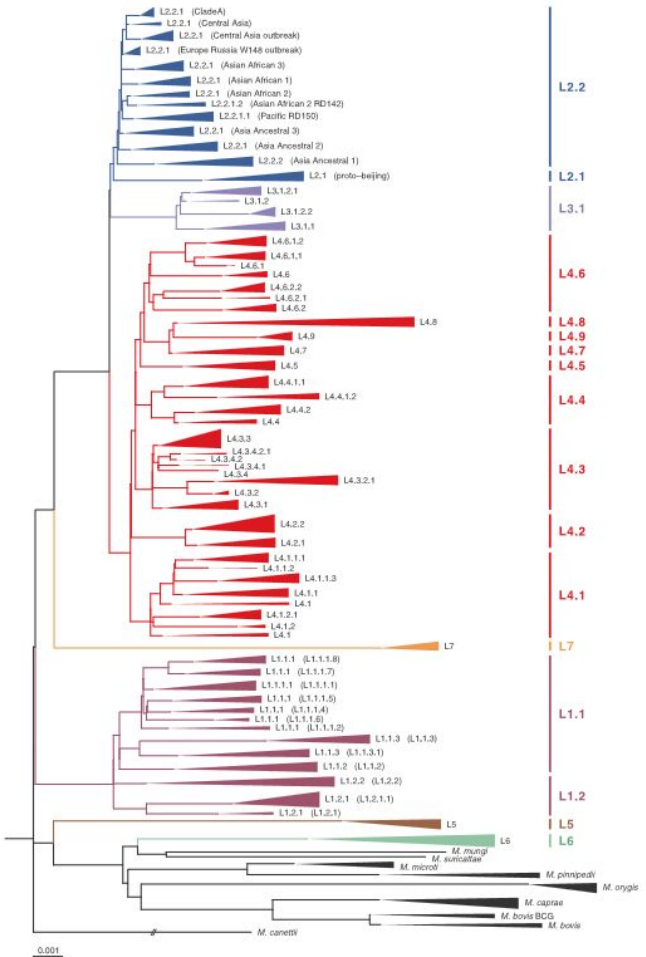
- evolutionary lineage of specific TB strain

Representation:

- 1D binary vector, N X N matrix, tree structure

Notes:

- Literature is split on using lineage as its commonly a confounder
- I will ablate with and without lineage

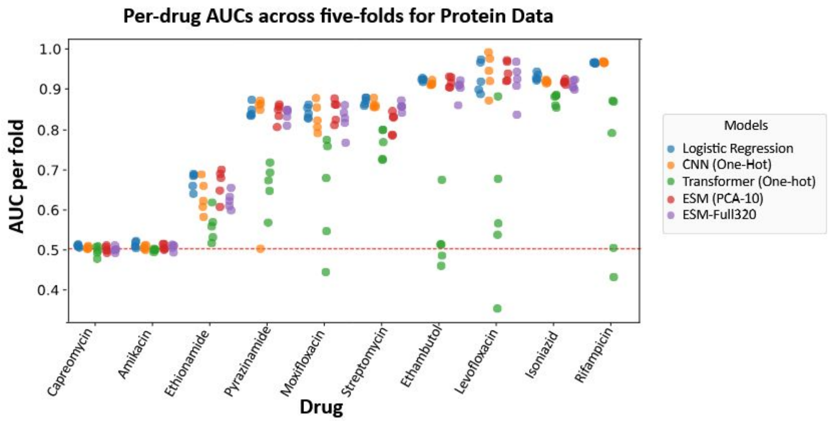


Model Design

Models - Literature

Trends:

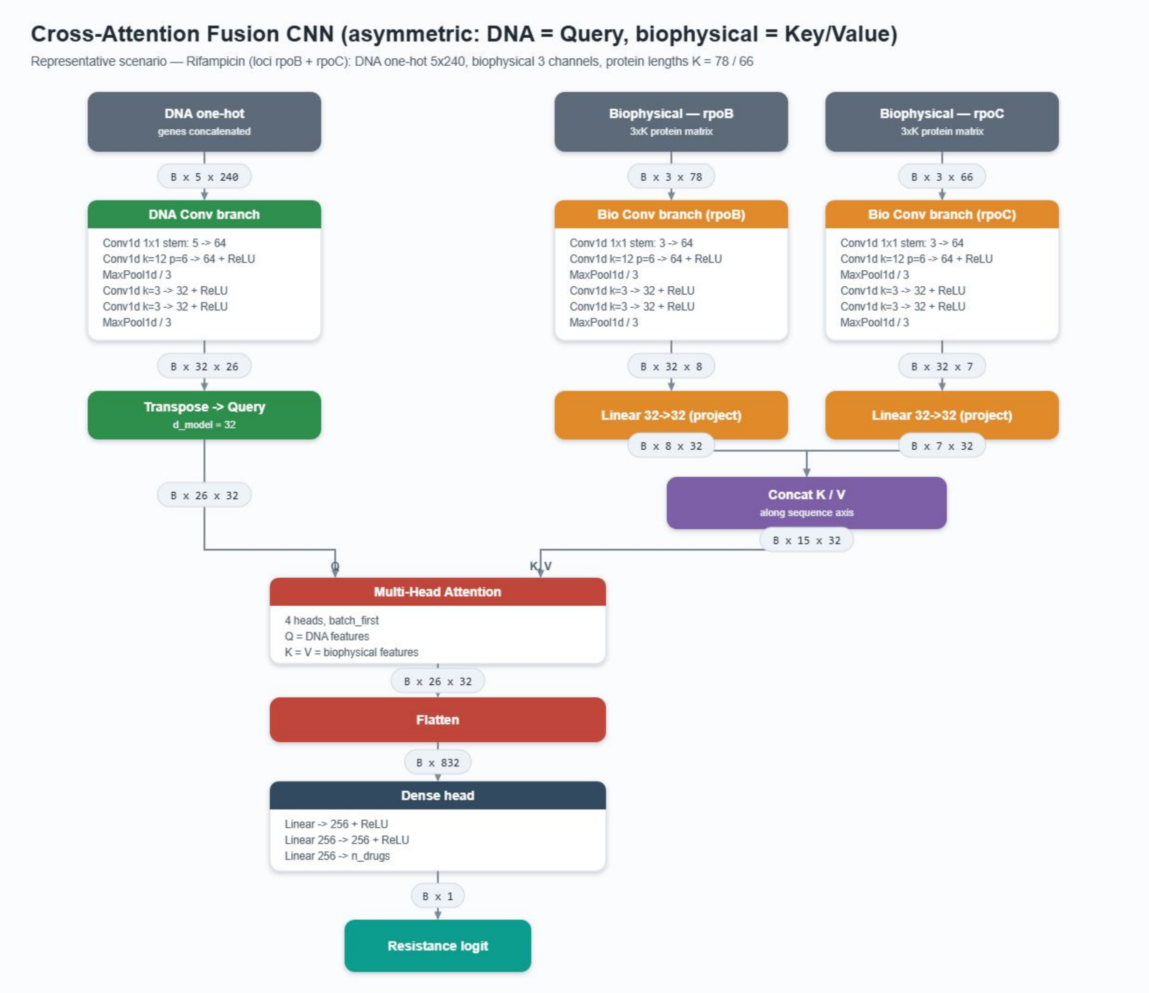
- CNN models consistently outperform other architectures
  - particularly on DNA sequence tasks
- DNA-sequence models consistently outperform protein-based models, likely because they capture crucial non-coding regulatory regions.
- Despite their complexity, language models struggle with second-line drugs and sequence-position averaging, largely due to data bottlenecks.



Models - Cross Attention

Description:

- to retain 1D CNN performance the model uses cross attention treating DNA as the query and other modalities as the key / value
- Should allow CNN insights to guide inference only retrieving information as needed



Models - Early Fusion

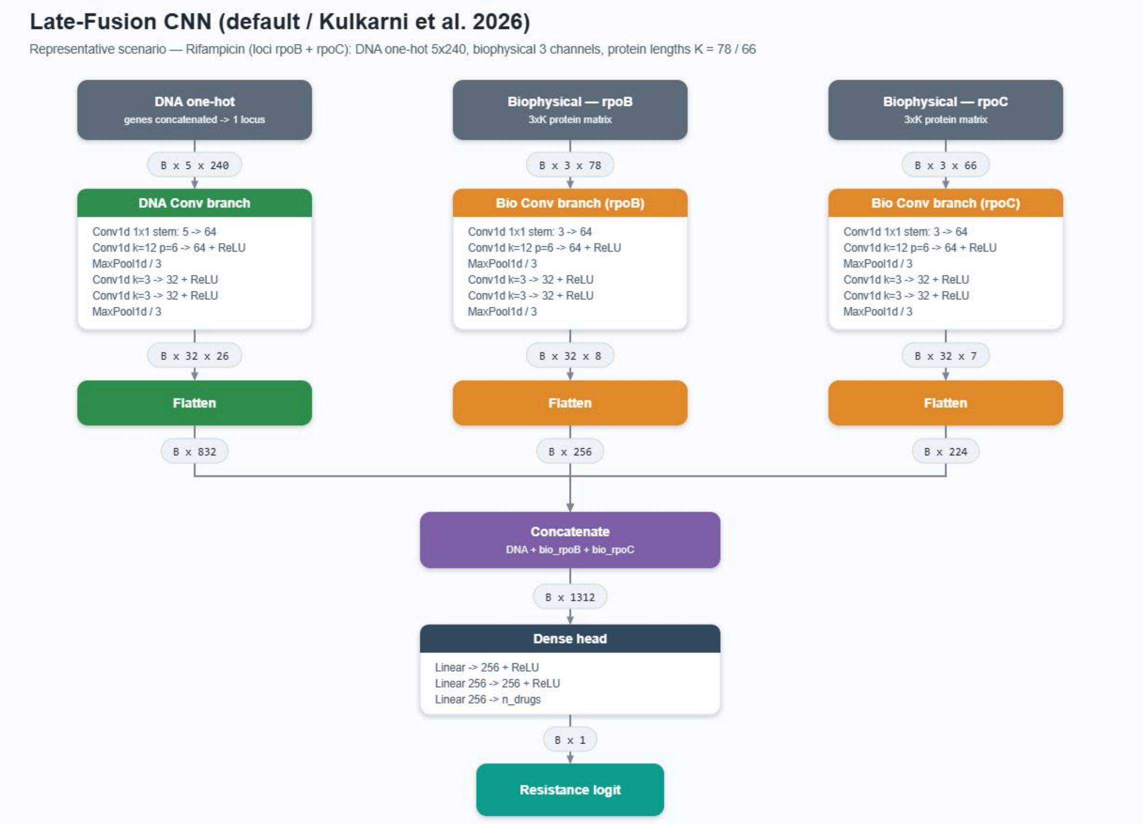
Description:

- concatenate all modalities into single vector
- pass concat vector into model as normal
- Fewer parameters and simpler implementation but lacking multi-modal nuance

Models - Late Fusion

Description:

- separate stacks for each modality
- allows different models to process different modalities
  - e.g. CNN for DNA and transformer for 3D distance
- requires more parameters but more robust
- can be trained with modality drop-out



Technique Ideas

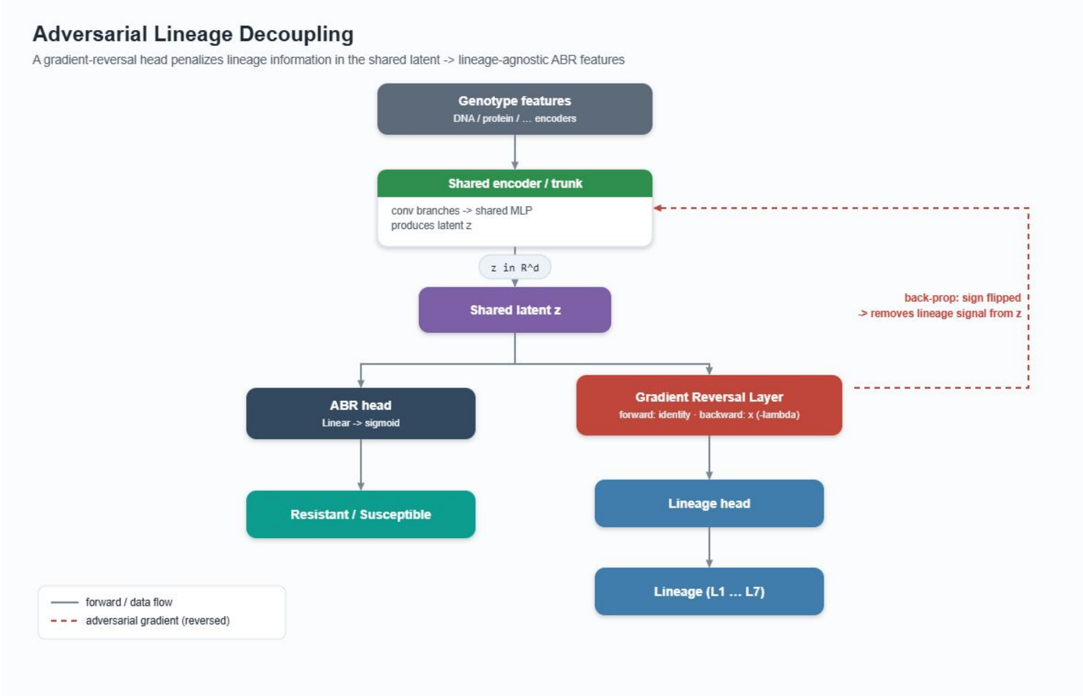
Techniques - Adversarial Lineage Decoupling

Description:

- second head predicts lineage off the shared latent
- its gradient is sign-flipped so the latent becomes lineage-agnostic

Notes:

- directly targets models over-relying on lineage
- keeps the ABR signal, removes population-structure signal



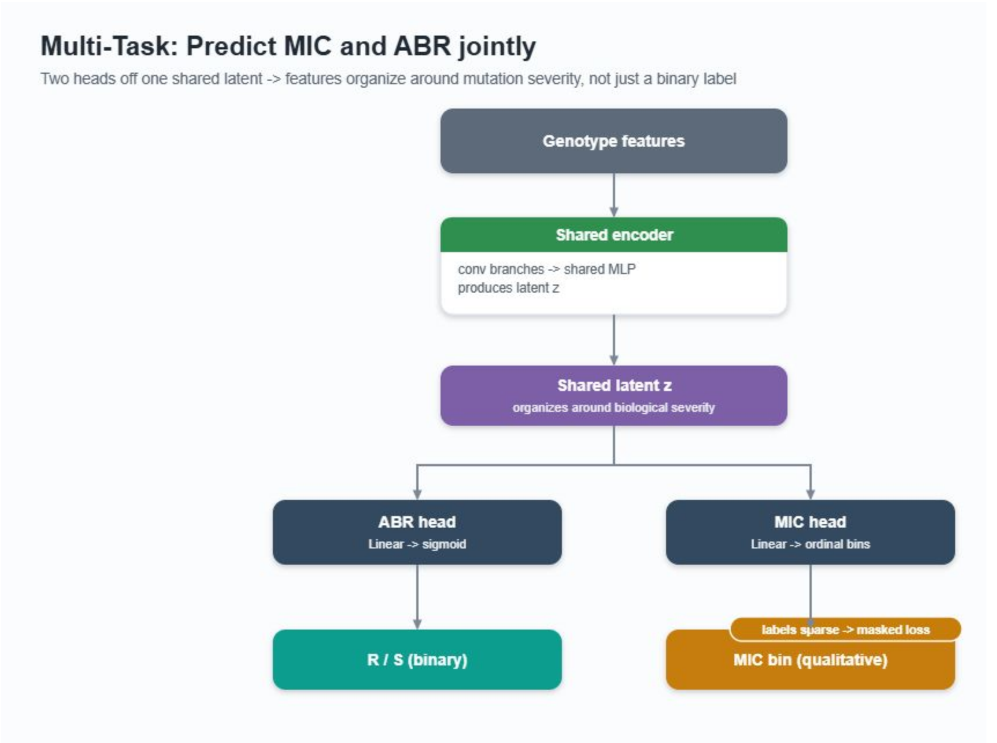
Techniques - Predict MIC and ABR (Multi-Task)

Description:

- add a second head predicting MIC alongside ABR, off the same latent
- forces the latent to organize around mutation severity, not just a threshold

Notes:

- MIC labels are sparse and qualitative -> mask the loss when absent



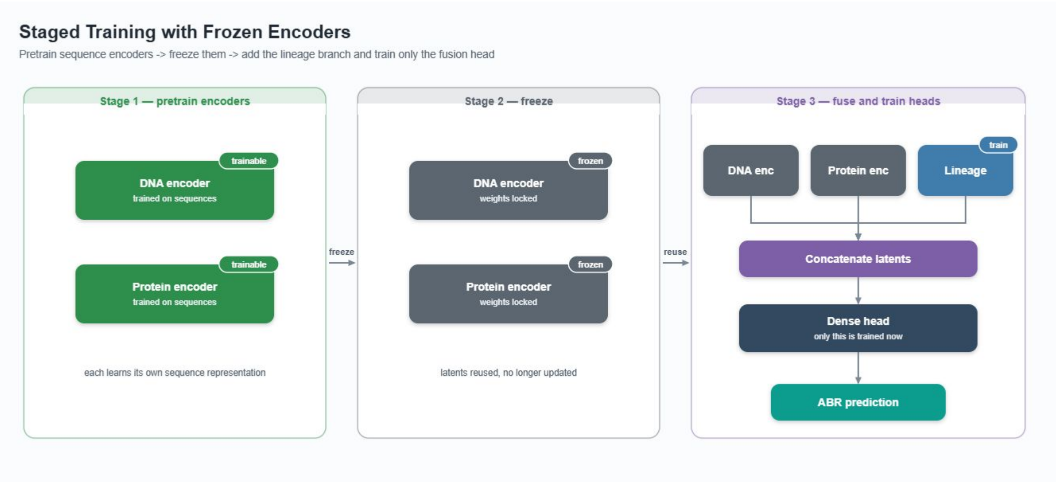
Techniques - Staged Training with Frozen Encoders

Description:

- 3 stages: pretrain DNA + protein encoders on sequence -> freeze -> add lineage branch, train only the fusion head
- fuses heterogeneous modalities without destabilizing pretrained encoders

Notes:

- cheap way to add lineage late and control its influence
- pairs with modality drop-out

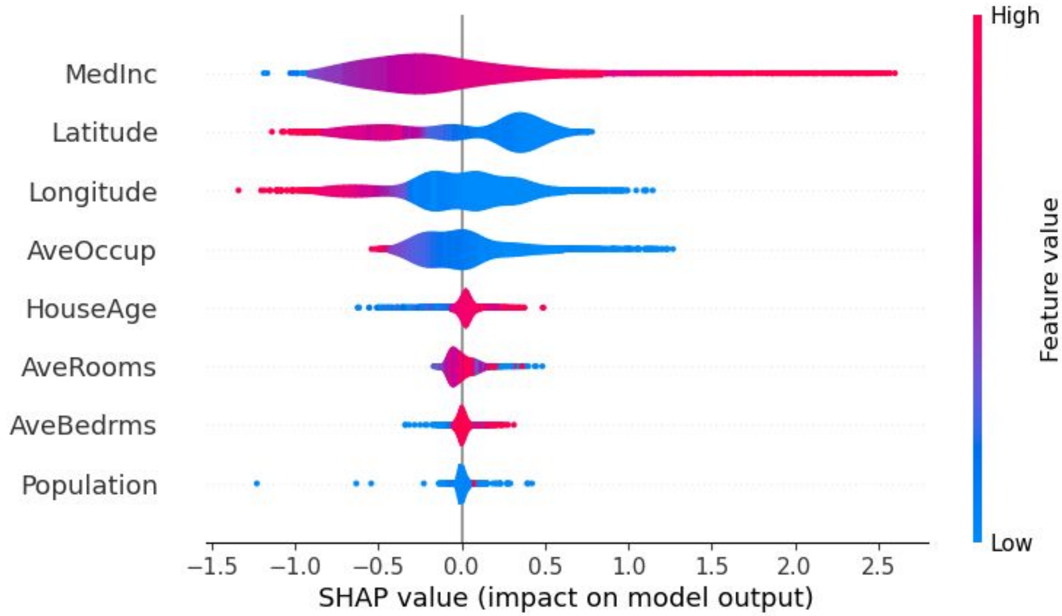


Interpretability

Interp - SHAP

Description:

- Open Source
- mechanistic interpretability library for input attribution
- commonly used in Comp Bio papers



Drawbacks:

- no novelty
- limited insights

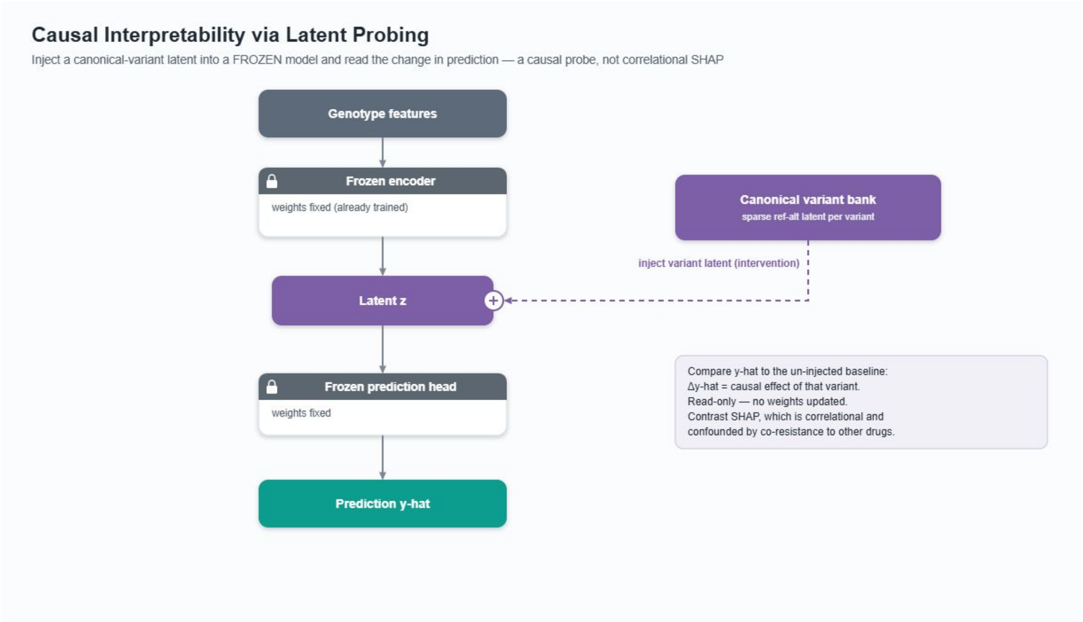
Interp - Causal Intervention

Description:

- inject canonical ref-alt variant latents into a frozen model, read change in prediction
- causal effect of a variant, vs correlational SHAP

Notes:

- diagnostic only, no performance gain
- avoids the common confounder (co-resistance to other drugs)



Summary

Summary - Big Picture

Modalities:

- DNA Sequence
    - regulatory regions
- Protein Sequence
- 3D Structure
- Molecular Characteristics
- Lineage

Models:

- Cross attention from DNA -> other modalities
- Late fusion with independent encoder stacks
- early fusion

Techniques:

- Adversarial Lineage Decoupling
- Multi-task prediction
- Staged training with frozen encoders

Interpretability:

- Start using SHAP
- design causal interventions by inserting reference latents into model during inference

Summary - Experiments

Planned Experiments:

Modality Zoo:

- 1-hot DNA sequence
- 1-hot regulatory regions
- 5 X k molecular characteristics

Model Zoo:

- Cross attention (DNA -> regulatory and MC)
- Late fusion (CNN for DNA and REG, Transformer for MC)
- Early fusion (all concat into one vector)

21 total permutations with modality ablation on each model choice. results will inform modality and model choice for further refinement and interpretability experiments.

Follow-up:

- include lineage / un-learn lineage
- explore transformer vs cnn per modality

Summary - Conclusion

Update Since Last Meeting:

- copied Big-TB code to Unity workspace
- code implementation
  - models
  - data loaders

Blockers:

- I need access to *pi_annagreen_umass_edu*
  - all data is stored there

For Next Meeting:

- replicate experiments with 1D CNN uni-modal single drug prediction
- validate pipeline with real data
- clean and prepare DNA sequence, regulatory regions, and MC feature data for inference